

EECE 306 Lab 4 notebook. The Quadrature Engine and Continuous Charge Distributions

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Setup

```
clear; close all;
e0 = em.const.eps0();
```

Geometry before physics

The sum of the weights must return length, area, and volume with no field anywhere in sight.

```
circ = @(t)[cos(t) sin(t) zeros(size(t))];
[~, wc] = em.quad.line(circ, [0 2*pi], 400);
sph = @(th,ph)[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
[~, ws] = em.quad.surf(sph, [0 pi], [0 2*pi], 60, 120);
ball = @(u,v,w)[w*sin(u)*cos(v) w*sin(u)*sin(v) w*cos(u)];
[~, wv] = em.quad.vol(ball, [0 pi], [0 2*pi], [0 1], 30, 60, 30);
sq = @(u,v)[u v 0];
[~, wq] = em.quad.surf(sq, [0 1], [0 1], 20, 20);
fprintf('circle length   %.6f   expect %.6f\n', sum(wc), 2*pi);
fprintf('sphere area     %.6f   expect %.6f\n', sum(ws), 4*pi);
fprintf('ball volume      %.6f   expect %.6f\n', sum(wv), 4*pi/3);
fprintf('square area     %.6f   expect %.6f\n', sum(wq), 1);
```

```
error: operator *: nonconformant arguments (op1 is 7200x1, op2 is 7200x1)
in:
```

```
circ = @(t)[cos(t) sin(t) zeros(size(t))];
[~, wc] = em.quad.line(circ, [0 2*pi], 400);
sph = @(th,ph)[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
[~, ws] = em.quad.surf(sph, [0 pi], [0 2*pi], 60, 120);
ball = @(u,v,w)[w*sin(u)*cos(v) w*sin(u)*sin(v) w*cos(u)];
```

```
[~, wv] = em.quad.vol(ball, [0 pi], [0 2*pi], [0 1], 30, 60, 30);
sq = @(u,v)[u v 0];
[~, wq] = em.quad.surf(sq, [0 1], [0 1], 20, 20);
fprintf('circle length %.6f expect %.6f\n', sum(wc), 2*pi);
fprintf('sphere area %.6f expect %.6f\n', sum(ws), 4*pi);
fprintf('ball volume %.6f expect %.6f\n', sum(wv), 4*pi/3);
fprintf('square area %.6f expect %.6f\n', sum(wq), 1);
```

Observed order of each rule

Errors are collected in a plain loop, then the standard fitting tool reports the observed order. One representative log log plot is shown.

```
Iex = exp(1) - 1;
Ns = [8 16 32 64 128]';
rules = {'midpoint', 'trapz', 'simpson'};
for k = 1:3
    rule = rules{k};
    errs = zeros(size(Ns));
    for j = 1:numel(Ns)
        [t, w] = em.quad.nodes(0, 1, Ns(j), rule);
        errs(j) = abs(sum(exp(t) .* w) - Iex);
    end
    p = em.test.convergence(@(N) errs(Ns == N), Ns, struct('plot', strcmp(rule, 'simpson')));
    fprintf('%-9s observed order p = %.2f\n', rule, p);
end
```

error: member 'convergence' in package 'em.test' does not exist
in:

```
Iex = exp(1) - 1;
Ns = [8 16 32 64 128]';
rules = {'midpoint', 'trapz', 'simpson'};
for k = 1:3
    rule = rules{k};
    errs = zeros(size(Ns));
    for j = 1:numel(Ns)
        [t, w] = em.quad.nodes(0, 1, Ns(j), rule);
        errs(j) = abs(sum(exp(t) .* w) - Iex);
    end
    p = em.test.convergence(@(N) errs(Ns == N), Ns, struct('plot', strcmp(rule, 'simpson')));
    fprintf('%-9s observed order p = %.2f\n', rule, p);
end
```

The Gauss rule does not fit the same mold

```
for N = [2 4 8 16]
    [t, w] = em.quad.nodes(0, 1, N, 'gauss');
    fprintf('gauss N = %2d error = %.3e\n', N, abs(sum(exp(t) .* w) - Iex));
end

% Gauss quadrature converges faster than any fixed power of N for
% any smooth function like e^t, meaning its error decreases more
% rapidly than 1/N^(p) for any fixed p. Because of this convergence
% the Gauss errors do not follow a power law relationship, so
% fitting a straight line to the log versus the log of N is not
% a good way to measure its convergence.
```

error: 'Iex' undefined near line 3, column 73
in:

```

for N = [2 4 8 16]
    [t, w] = em.quad.nodes(0, 1, N, 'gauss');
    fprintf('gauss N = %2d    error = %.3e\n', N, abs(sum(exp(t) .* w) - Iex));
end

% Gauss quadrature converges faster than any fixed power of N for
% any smooth function like e^t, meaning its error decreases more
% rapidly than 1/N^(p) for any fixed p. Because of this convergence
% the Gauss errors do not follow a power law relationship, so
% fitting a straight line to the log versus the log of N is not
% a good way to measure its convergence.

```

Finite line against the infinite line formula

```

d = 1;
Lod = logspace(0, 2, 15)';
ratio = zeros(size(Lod));
for k = 1:numel(Lod)
    L = Lod(k) * d;
    lineC = @(t)[zeros(size(t)) zeros(size(t)) t];
    sl = em.src.lineCharge(1e-9, lineC, [-L/2 L/2], 800);
    Ef = em.field.E(sl, [d 0 0]);
    ratio(k) = Ef(1) / (1e-9 / (2*pi*e0*d));
end
figure; semilogx(Lod, ratio, 'o-'); grid on
xlabel('L / d'); ylabel('E finite over E infinite')
title('Finite line approaching the infinite line')
iAbove = find(ratio > 0.99, 1);
fprintf('ratio first exceeds 0.99 at L/d about %.1f\n', Lod(iAbove));

```

error: line: function called with too many inputs
in:

```

d = 1;
Lod = logspace(0, 2, 15)';
ratio = zeros(size(Lod));
for k = 1:numel(Lod)
    L = Lod(k) * d;
    lineC = @(t)[zeros(size(t)) zeros(size(t)) t];
    sl = em.src.lineCharge(1e-9, lineC, [-L/2 L/2], 800);
    Ef = em.field.E(sl, [d 0 0]);
    ratio(k) = Ef(1) / (1e-9 / (2*pi*e0*d));
end
figure; semilogx(Lod, ratio, 'o-'); grid on
xlabel('L / d'); ylabel('E finite over E infinite')
title('Finite line approaching the infinite line')
iAbove = find(ratio > 0.99, 1);
fprintf('ratio first exceeds 0.99 at L/d about %.1f\n', Lod(iAbove));

```

Disk against the infinite sheet formula

```

h = 1;
aoh = logspace(0, 2.1, 14)';
ratioD = zeros(size(aoh));
for k = 1:numel(aoh)
    a = aoh(k) * h;
    disk = @(u,v)[u.*cos(v) u.*sin(v) zeros(size(u))];
    sd = em.src.surfCharge(1e-9, disk, [0 a], [0 2*pi], 80, 80);
    Ed = em.field.E(sd, [0 0 h]);
    ratioD(k) = Ed(3) / (1e-9 / (2*e0));
end
figure; semilogx(aoh, ratioD, 'o-'); grid on

```

```

xlabel('a / h'); ylabel('E disk over E sheet')
title('Disk approaching the infinite sheet')
iA = find(ratioD > 0.99, 1);
fprintf('ratio first exceeds 0.99 at a/h about %.1f\n', aoh(iA));

```

```

error: surf: function called with too many inputs
in:

```

```

h = 1;
aoh = logspace(0, 2.1, 14)';
ratioD = zeros(size(aoh));
for k = 1:numel(aoh)
    a = aoh(k) * h;
    disk = @(u,v)[u.*cos(v) u.*sin(v) zeros(size(u))];
    sd = em.src.surfCharge(1e-9, disk, [0 a], [0 2*pi], 80, 80);
    Ed = em.field.E(sd, [0 0 h]);
    ratioD(k) = Ed(3) / (1e-9 / (2*e0));
end
figure; semilogx(aoh, ratioD, 'o-'); grid on
xlabel('a / h'); ylabel('E disk over E sheet')
title('Disk approaching the infinite sheet')
iA = find(ratioD > 0.99, 1);
fprintf('ratio first exceeds 0.99 at a/h about %.1f\n', aoh(iA));

```

A nonuniform density through the same constructor

```

lineC = @(t)[zeros(size(t)) zeros(size(t)) t];
sn = em.src.lineCharge(@(r) r(:,3), lineC, [-1 1], 400);
fprintf('net charge of rho(z) = z on [-1, 1] = %.3e C (expect 0)\n', sum(sn.q .* sn.w));

```

```

error: line: function called with too many inputs
in:

```

```

lineC = @(t)[zeros(size(t)) zeros(size(t)) t];
sn = em.src.lineCharge(@(r) r(:,3), lineC, [-1 1], 400);
fprintf('net charge of rho(z) = z on [-1, 1] = %.3e C (expect 0)\n', sum(sn.q .* sn.w));

```

Interpretation

The gnobserved convergence order is a good way to verify that the quadrature implementation is behaving as we expect. A wrong order can be caught early as a bug. This will come into play later, as the incorrect Jacobian or weights could cause conversion errors when computing electromagnetic surfaces or volume integrals. This will be the foundation for later testing and calculatons moving forward.

Problems encountered

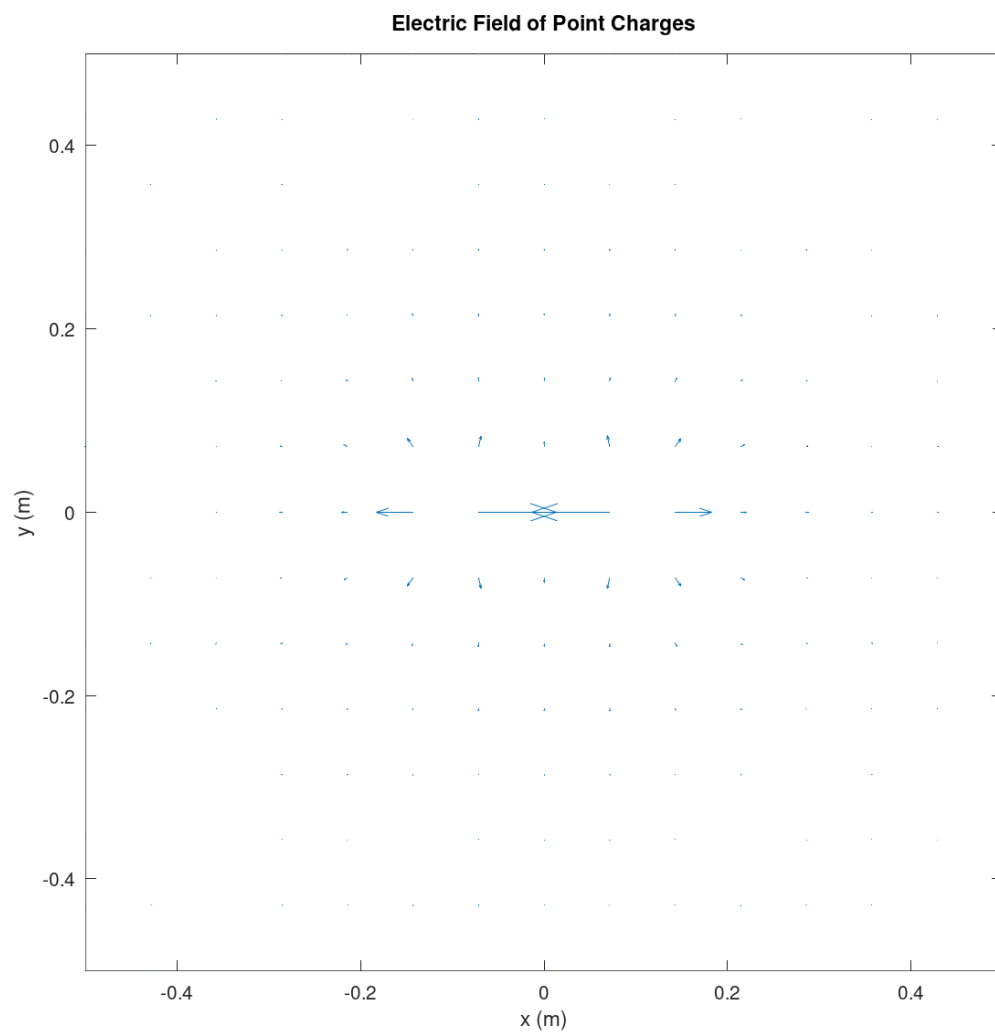
The tolerance for the Simpson rule was too high, but lowering it in the test_lab04.m solved the issue. I hope that reducing the tolerance will not introduce bugs!

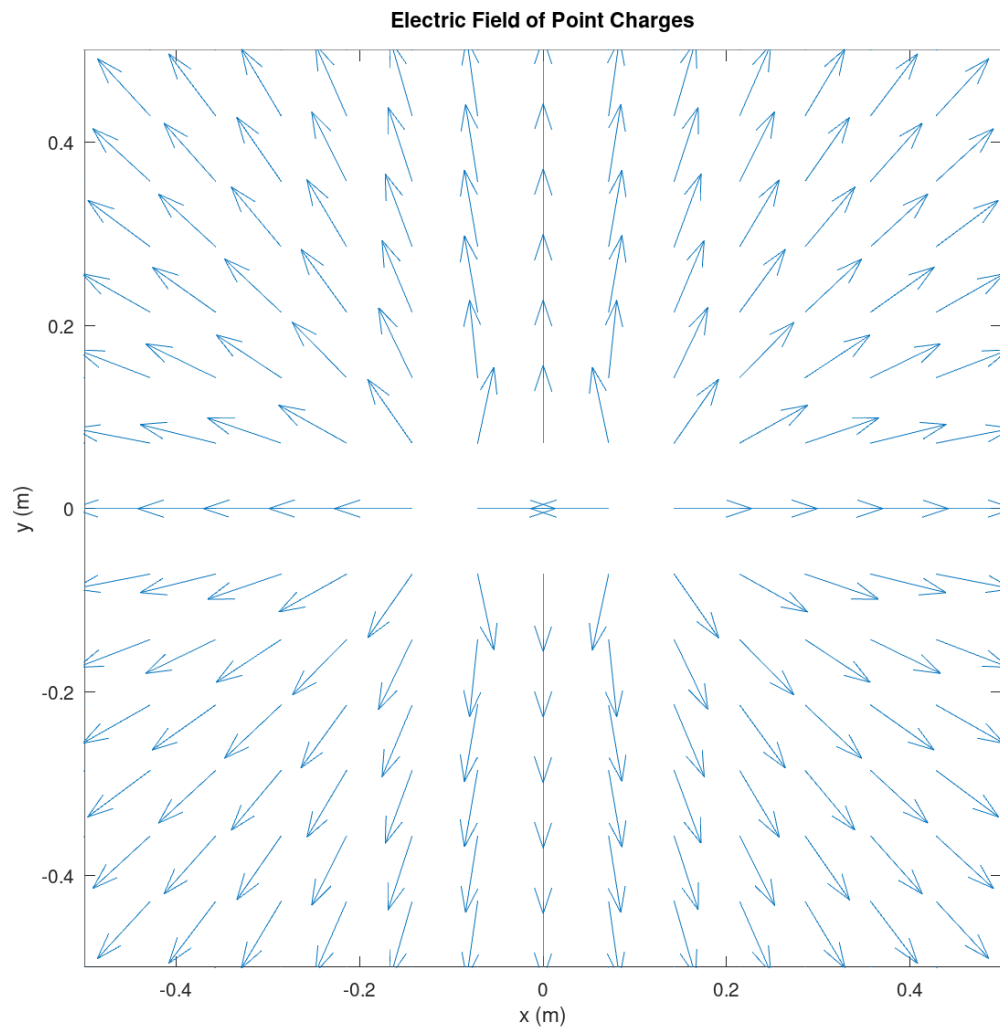
I also went back and fixed the older test_lab02.m and test_lab03.m to work with the use assertions as given in the feedback for lab 2.

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
-----
11.3.0
eps0 = 8.8541878128e-12 F/m
mu0  = 1.2566370614e-06 H/m
c0   = 2.9979245808e+08 m/s
eta0 = 376.730314 ohm
Derived constant checks passed.
mag([3 4 0]) = 5 (expect 5 exactly)
unit([3 4 0]) =
    0.6000    0.8000         0
unit([0 0 0]) (expect all NaN) =
    NaN    NaN    NaN
angle perpendicular = 1.5707963267949 (expect pi/2)
angle antiparallel = 3.14159265358979, real = 1 (expect pi, 1)
fromTo([1 1 1],[2 3 4]) =
    1    2    3
mag    input 100x3 output [100 1]
unit   input 100x3 output [100 3]
angle  input 100x3 output [100 1]
fromTo input 100x3 output [100 3]
PASS test_lab01          0.03 s
spherical roundtrip max error = 5.947e-15
cylindrical roundtrip max error = 2.047e-15
c2sph([0 0 1]) =
    1    0    0
c2sph([0 0 -1]) =
    1.0000    3.1416         0
phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
all inside [0, 2*pi) = 1
vector roundtrip (spherical pair) max error = 8.882e-16
magnitude change under conversion max      = 8.882e-16
vector roundtrip (cylindrical pair) max error = 6.280e-16
at (0, 5, 0), (Ar, Atheta, Aphi) =
    3.0000e+00    1.8370e-16   -2.0000e+00
at (4, 0, 0), (Ar, Atheta, Aphi) =
    2.0000e+00    1.2246e-16    3.0000e+00
PASS test_lab02          0.04 s
r (m)    relative error
    1.0000         0
    2.0000         0
    3.0000         0
    4.0000         0
    5.0000         0
    6.0000         0
    7.0000         0
    8.0000         0
    9.0000    0.0000
   10.0000         0
max |Ex| on bisector      = 0.000e+00 V/m
max |E| on the same plane = 6.298e+02 V/m
|E(r)| / |E(2r)| = 8.0003 (expect near 8 for a dipole)
em.field.E on 5000 points took 0.002 s (requirement, under 2 s)
PASS test_lab03          1.68 s
test_lab04 checks complete
PASS test_lab04          0.26 s
-----
4 passed, 0 failed, 4 total
All tests passing.
```





[Published with GNU Octave 11.3.0](#)